

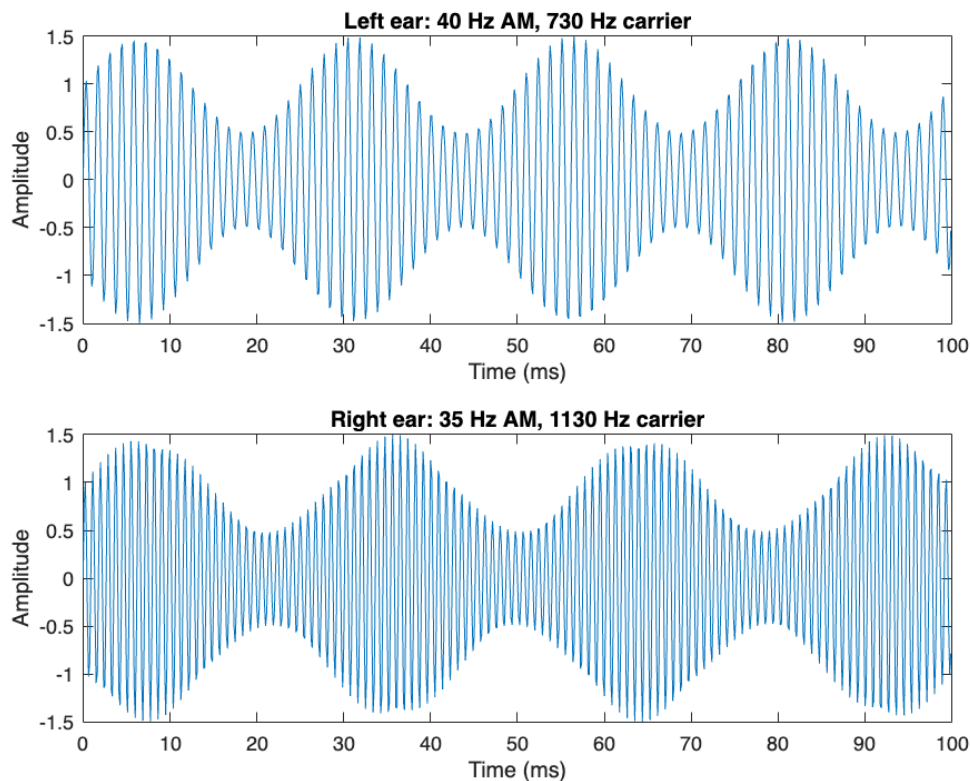
Plot the time waveform of the sounds used in your experiments (only the first 100ms of the signal, separately for the left ear and the right ear) and their time-frequency decomposition called a spectrogram (use specgram function in MATLAB). (1 pt)

```
fs = 8000; dur = 0.1; t = 0:1/fs:dur-1/fs;

% 40 Hz modulated @ 730 Hz left n 35 Hz modulated @ 1130 Hz right
left = (1 + 0.5*sin(2*pi*40*t)) .* sin(2*pi*730*t);
right = (1 + 0.5*sin(2*pi*35*t)) .* sin(2*pi*1130*t);

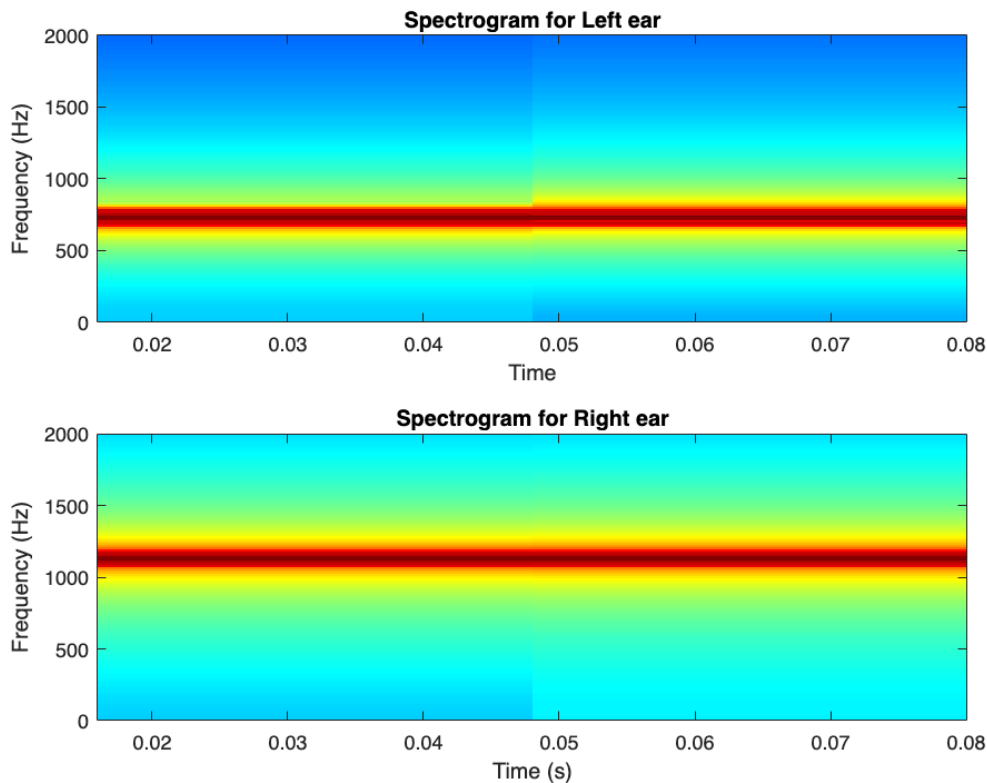
% the waveforms that were generated
figure('Color','w');
subplot(2,1,1);
plot(t*1000, left); xlim([0 100]);
xlabel('Time (ms)'); ylabel('Amplitude');
title('Left ear: 40 Hz AM, 730 Hz carrier');

subplot(2,1,2);
plot(t*1000, right); xlim([0 100]);
xlabel('Time (ms)'); ylabel('Amplitude');
title('Right ear: 35 Hz AM, 1130 Hz carrier');
```



```
% our spectrograms, could zoom in to see the 730 +- 40 and 1130 +- 35
figure('Color','w');
subplot(2,1,1);
specgram(left, 512, fs); ylim([0 2000]);
title('Spectrogram for Left ear'); ylabel('Frequency (Hz)'); colormap jet;

subplot(2,1,2);
specgram(right, 512, fs); ylim([0 2000]);
title('Spectrogram for Right ear'); ylabel('Frequency (Hz)'); xlabel('Time
(s)'); colormap jet;
```



Load the EEG signal in Matlab. Use the trigger signal from the s-function to separate the EEG response to sound. (1 pt)

```
%trigger has to be equal to 1 (just getting rid of the first 15s)
files = {
    'lab7_exp1_trial1_attend_left.mat'
    'lab7_exp1_trial2_attend_right.mat'
    'lab7_exp1_trial3_attend_both.mat'
    'lab7_exp1_trial4_unattended.mat'};

% make sure to have the correct mappings
chanIDs = [1 2 3 6 7 8 11];
```

```

chanLabs = {'Fz', 'FC2', 'FC1', 'Cz', 'CP1', 'CP2', 'Pz'};

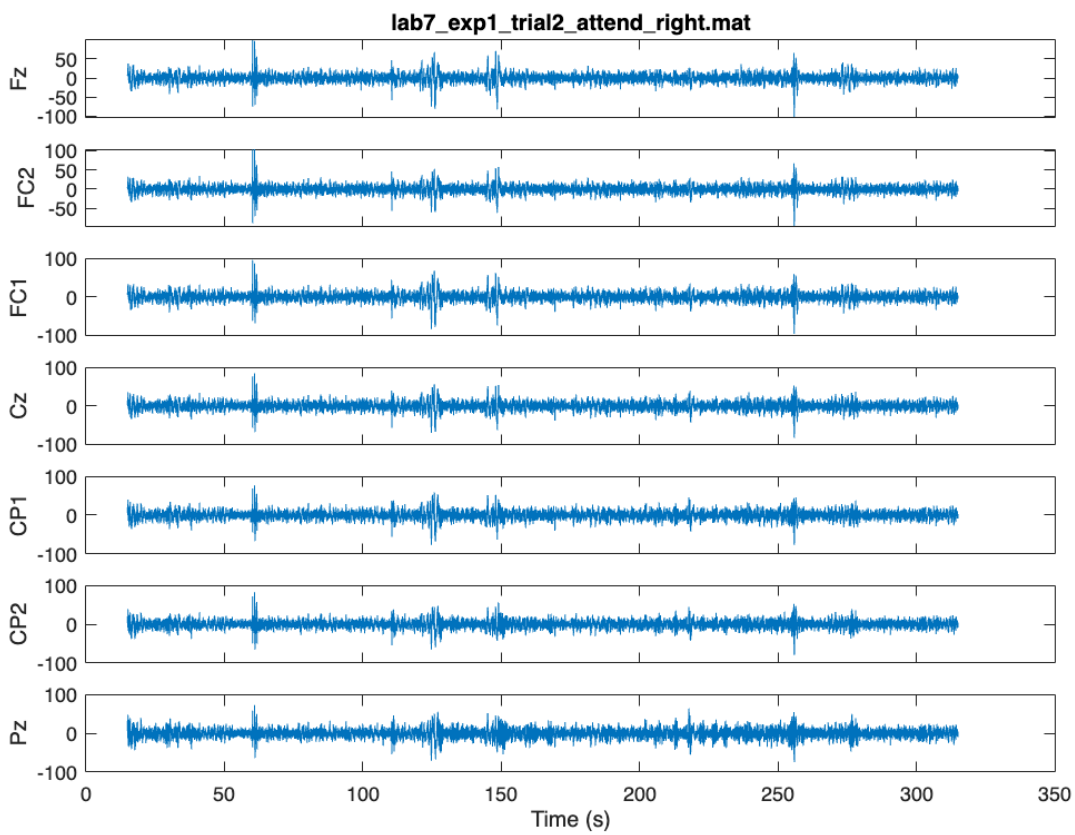
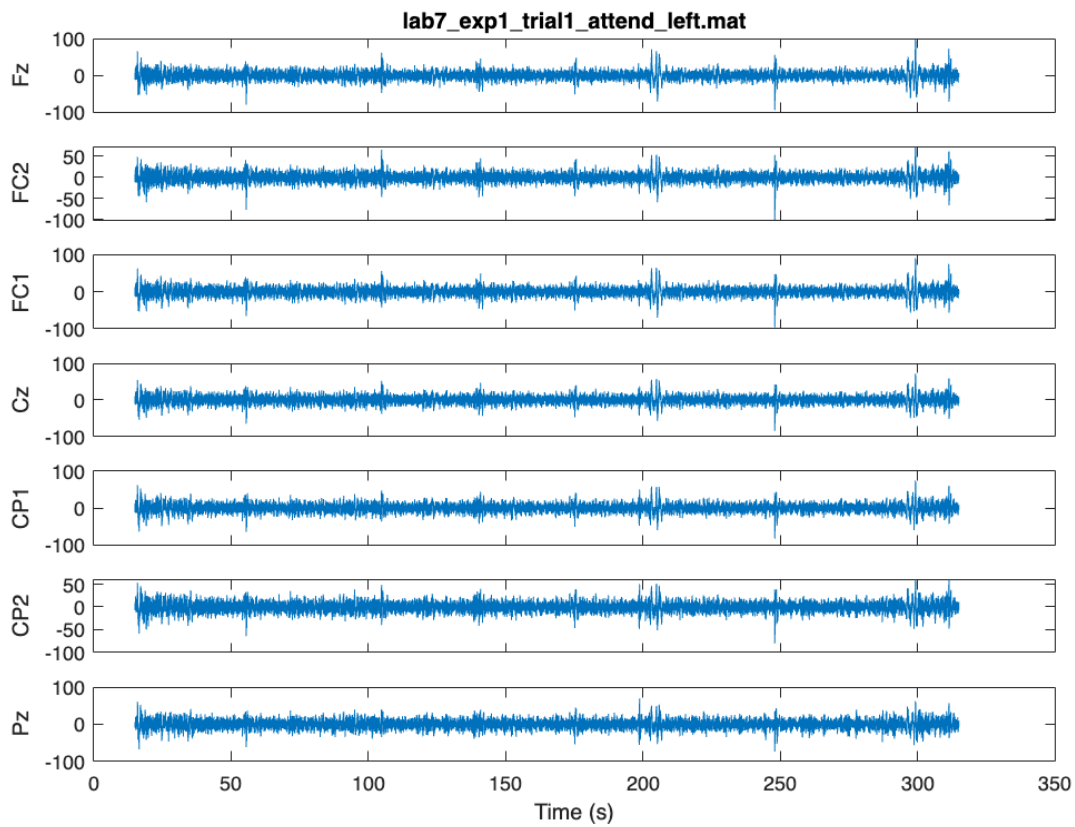
for f = 1:numel(files)
    S = load(files{f}); %since init stored as trigger.
    ts = S.trigger;
    t = ts.Time(:);
    X = ts.Data; %n x 8 with the last col trigger (1 or 0)
    trig = X(:,end);
    mask = trig == 1;

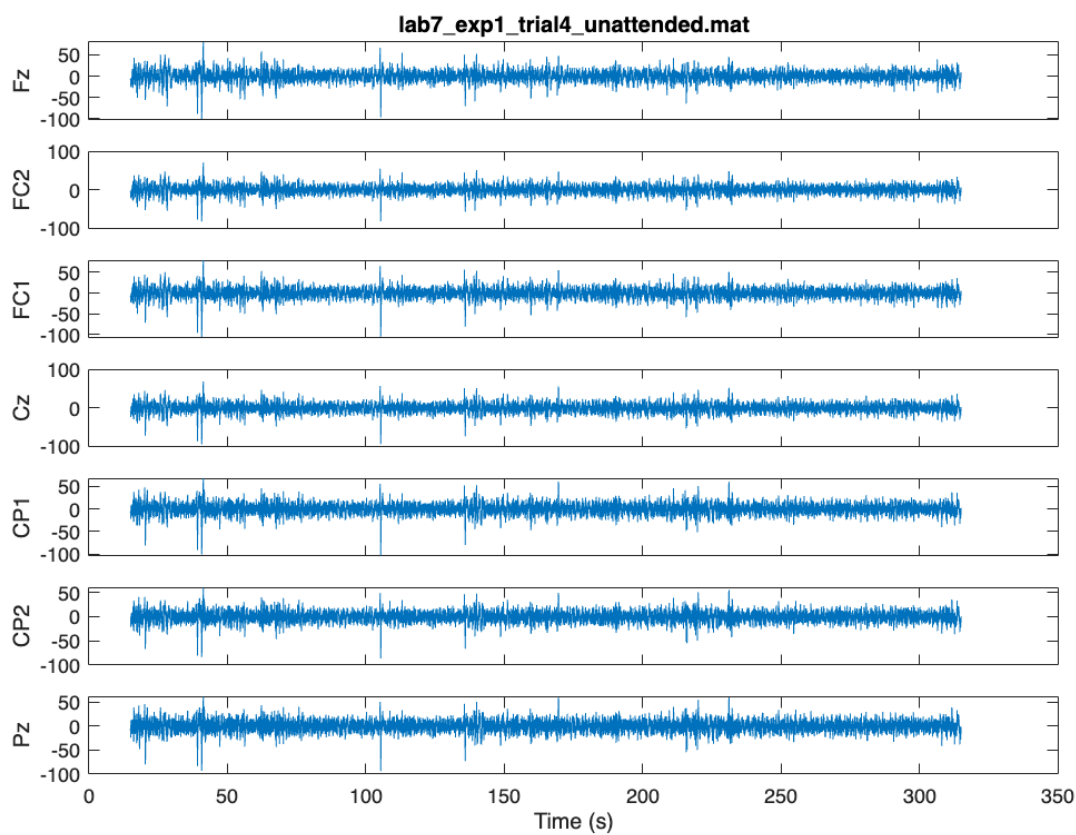
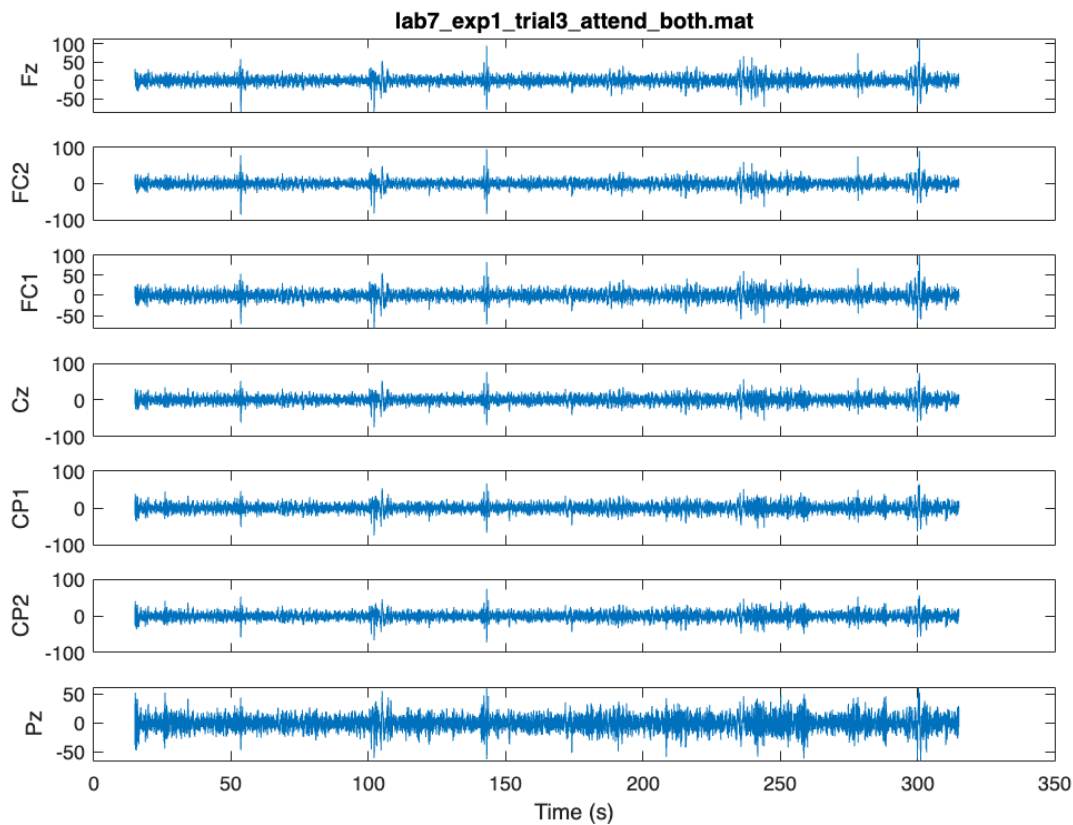
    % cols in asc order before trigger
    id2col = containers.Map(num2cell(chanIDs), num2cell(1:numel(chanIDs)));

    % plot some channels before stimulus
    figure('Color','w','Name',['EEG (trigger=1) - ' files{f}]);
    tl =
    tiledlayout(numel(chanIDs),1,'TileSpacing','compact','Padding','compact');

    for k = 1:numel(chanIDs)
        c = id2col(chanIDs(k)); % col index only 1-7, 8 trigger
        nexttile;
        plot(t(mask), X(mask, c));
        ylabel(chanLabs{k});
        if k==1, title(strrep(files{f},'_','\_')); end
        if k==numel(chanIDs), xlabel('Time (s)'); else,
set(gca,'XTickLabel',[]); end
    end
end

```





Take the Fourier transform of each channel, and then average the magnitude

of FFT over all channels. (1 pt)

```
%% Full + Zoomed FFT per file (trigger == 1, wide layout, faint markers)

fs = 352;    % EEG sampling rate (Hz)

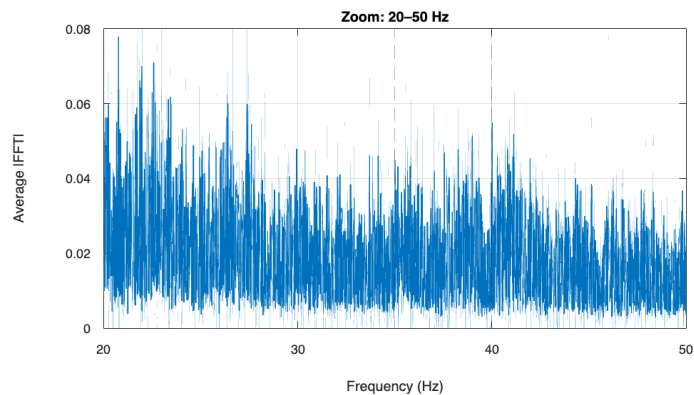
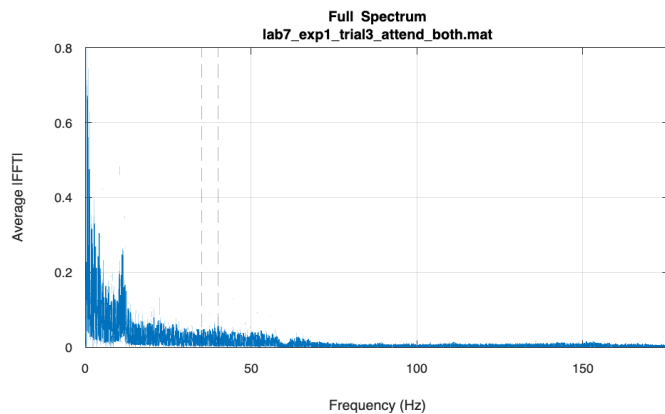
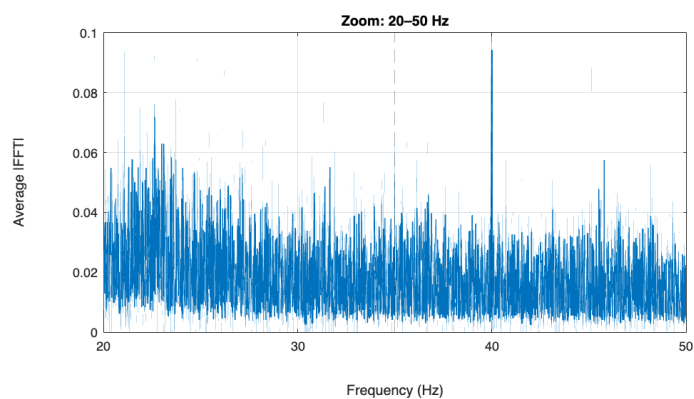
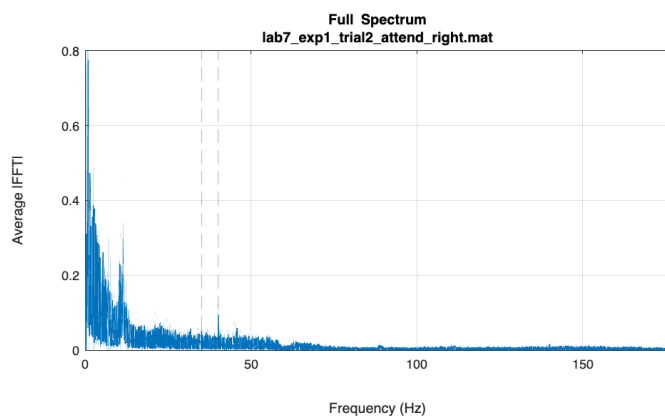
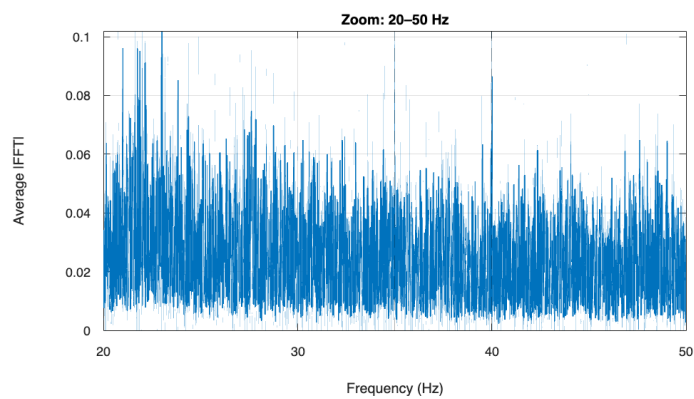
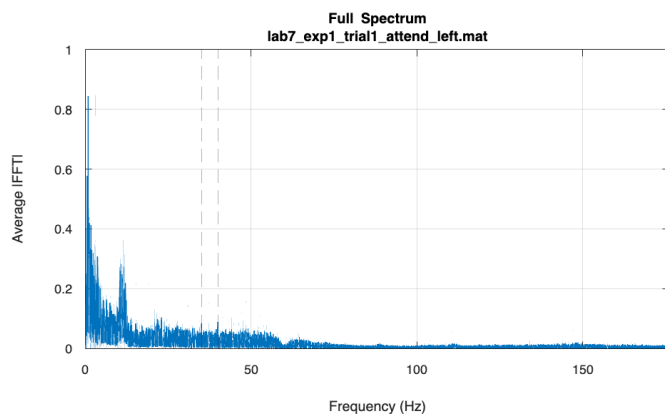
for f = 1:numel(files)
    S = load(files{f});
    X = S.trigger.Data;
    trig = X(:,end);
    X_stim = X(trig==1, 1:numel(chanIDs));    % EEG only while trigger ==
1
    N = size(X_stim,1);
    if N < 2, warning('%s: no valid samples', files{f}); continue; end

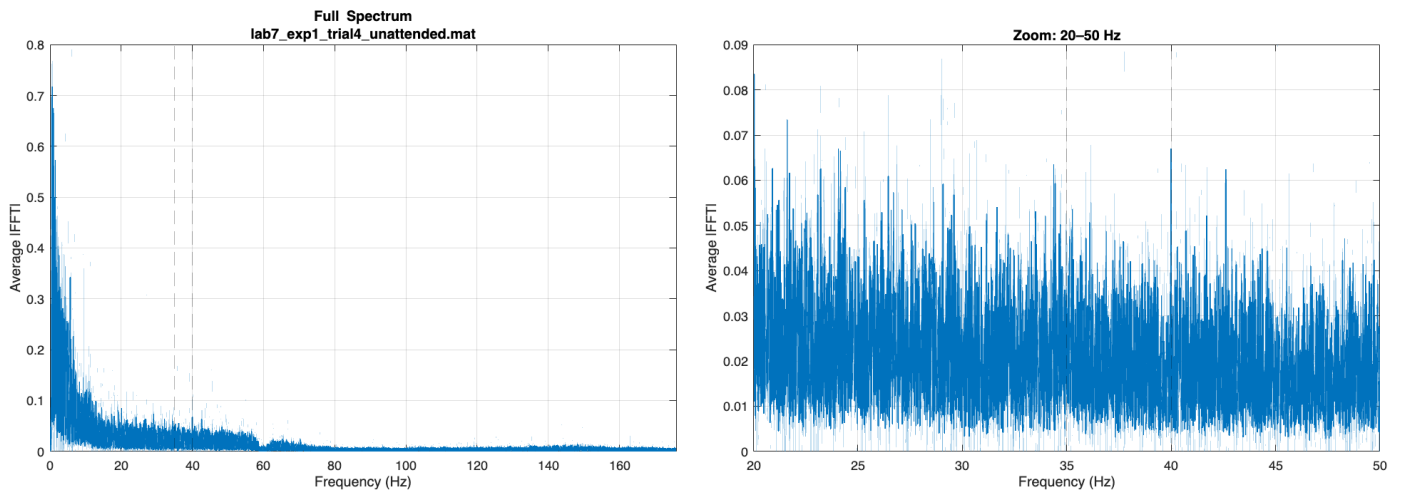
    % Detrend + window
    Xw = (X_stim - mean(X_stim,1)) .* hann(N);
    Y = abs(fft(Xw)/N);
    P1 = 2*Y(1:floor(N/2)+1, :);
    avg_fft = mean(P1, 2);
    f_axis = (0:floor(N/2))*(fs/N);

    % --- Create wider figure ---
    fig = figure('Color','w','Name',files{f});
    fig.Position(3:4) = [1200 400];    % width=1200px, height=400px
    tiledlayout(1,2,'TileSpacing','compact','Padding','compact');

    % --- Full spectrum (left) ---
    nexttile;
    plot(f_axis, avg_fft, 'LineWidth',1.2); hold on;
    xline([35 40],'--k','LineWidth',0.5,'Alpha',0.3);
    hold off; grid on;
    xlabel('Frequency (Hz)'); ylabel('Average |FFT|');
    title({'Full Spectrum', strrep(files{f},'_','\_')});
    xlim([0 fs/2]);

    % --- Zoomed 20-50 Hz (right) ---
    nexttile;
    plot(f_axis, avg_fft, 'LineWidth',1.2); hold on;
    xline([35 40],'--k','LineWidth',0.5,'Alpha',0.3);
    hold off; grid on;
    xlabel('Frequency (Hz)'); ylabel('Average |FFT|');
    title('Zoom: 20-50 Hz');
    xlim([20 50]);
end
```





Question: if the length of the FFT is L , and given the sampling frequency of your EEG signal (256Hz), what samples in FFT correspond to the modulator frequencies of the left ear and the right ear? (i.e. 40Hz & 35Hz)? (1 pt)

```

%% lets do it for every file just to make sure
fs = 352; %Hz problem stil persists

for f = 1:numel(files)
    S = load(files{f});
    X = S.trigger.Data;
    trig = X(:,end);

    % of course only count samples where the trigger equal 1
    N = sum(trig == 1);

    % compute the fft bins and add 1 bc MATLAB
    bin35 = round(35 * N / fs) + 1;
    bin40 = round(40 * N / fs) + 1;

    fprintf('\nFile: %s\n', files{f});
    fprintf('FFT bin for 35 Hz = %d\n', bin35);
    fprintf('FFT bin for 40 Hz = %d\n', bin40);

    % and the freq resolution
    df = fs / N;
    fprintf('Each FFT bin represents %.6f Hz\n', df);
    % lets convert the bins back into their frequency just as a sanity
    % check
    f_from_35 = (bin35 - 1) * df;
    f_from_40 = (bin40 - 1) * df;
    fprintf('Recovered frequencies are: 35 Hz → %.4f Hz, 40 Hz → %.4f
    Hz\n', f_from_35, f_from_40);
end

```



```

File: lab7_exp1_trial1_attend_left.mat
FFT bin for 35 Hz = 10501
FFT bin for 40 Hz = 12001
Each FFT bin represents 0.003333 Hz
Recovered frequencies are: 35 Hz → 34.9997 Hz, 40 Hz → 39.9996 Hz
File: lab7_exp1_trial2_attend_right.mat
FFT bin for 35 Hz = 10501
FFT bin for 40 Hz = 12001
Each FFT bin represents 0.003333 Hz
Recovered frequencies are: 35 Hz → 34.9997 Hz, 40 Hz → 39.9996 Hz
File: lab7_exp1_trial3_attend_both.mat
FFT bin for 35 Hz = 10501
FFT bin for 40 Hz = 12001
Each FFT bin represents 0.003333 Hz
Recovered frequencies are: 35 Hz → 34.9997 Hz, 40 Hz → 39.9996 Hz
File: lab7_exp1_trial4_unattended.mat
FFT bin for 35 Hz = 10501
FFT bin for 40 Hz = 12001
Each FFT bin represents 0.003333 Hz
Recovered frequencies are: 35 Hz → 34.9997 Hz, 40 Hz → 39.9996 Hz

```

Create a table or bar graph showing the average magnitude of the ASSR signal over all channels for the two modulator frequencies and for each of the four listening conditions. How do you interpret the data? (6 pts)

```

fs = 352;

freqs = [35, 40]; % our mod. freqs
nCh = numel(chanIDs);
condLabels = strings(numel(files),1);
assrMat = zeros(numel(files), numel(freqs)); % rows are equal to our
conditions, where the cols are 35Hz 40Hz

for f = 1:numel(files)
    %stim. on
    S = load(files{f});
    ts = S.trigger;
    X = ts.Data;
    trig = X(:,end);
    mask = (trig == 1);

    X_stim = X(mask, 1:nCh);

    % same thing detrend and then window
    Xw = X_stim - mean(X_stim,1);
    win = hann(N);
    Xw = Xw .* win;

    %all the same, one sided fft
    npos = floor(N/2) + 1;

```

```

df = fs / N;

% get the magnitude spectrum per channel
mag = zeros(npos, nCh);
for k = 1:nCh
    Y = fft(Xw(:,k));% N
    P2 = abs(Y / N);
    P1 = P2(1:npos);
    if N > 2
        if mod(N,2)==0, P1(2:end-1) = 2*P1(2:end-1);
        else,          P1(2:end)   = 2*P1(2:end);
        end
    end
    mag(:,k) = P1;
end

% now take the average across all of the channels
avgMag = mean(mag, 2);

% should now pull the bins for 35 Hz and then for 40 Hz
idx = round(freqs * N / fs) + 1; % here we have 1-based indices for our
onesided spectrum
%idx = min(max(idx,1), npos); % don't really need, just in case

assrMat(f, :) = avgMag(idx).'; % store our 35Hz n 40Hz

% get our condition label from the filenames
name = lower(files{f});
if contains(name,'attend_left'),   condLabels(f) = "Attend Left";
elseif contains(name,'attend_right'), condLabels(f) = "Attend Right";
elseif contains(name,'attend_both'), condLabels(f) = "Attend Both";
elseif contains(name,'unattended'),  condLabels(f) = "Unattended";
else, condLabels(f) = sprintf("Cond %d", f);
end
end

% gen. table
T = table(condLabels, assrMat(:,1), assrMat(:,2), ...
    'VariableNames', {'Condition', 'ASSR_35Hz', 'ASSR_40Hz'});
disp(T);

```

Condition	ASSR_35Hz	ASSR_40Hz
"Attend Left"	0.080824	0.086337
"Attend Right"	0.050038	0.094037
"Attend Both"	0.039756	0.054886
"Unattended"	0.027632	0.066845

```

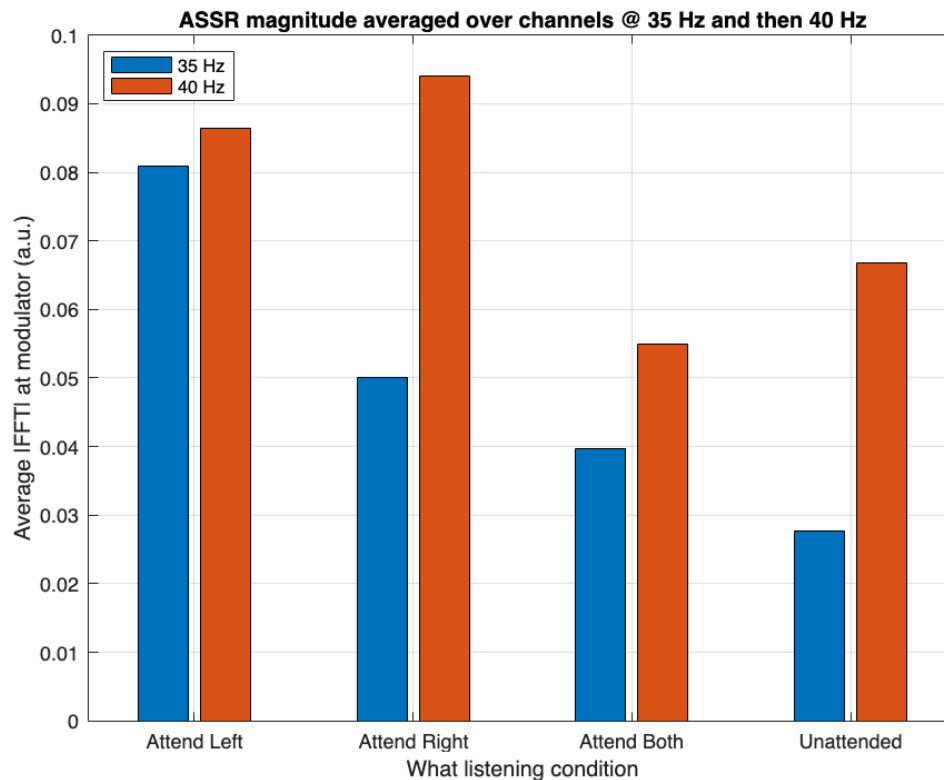
% gen. grouped bar plot
figure('Color','w');

```

```

bar(assrMat, 'grouped');
set(gca, 'XTick', 1:numel(files), 'XTickLabel', condLabels);
xlabel('What listening condition'); ylabel('Average |FFT| at modulator (a.u.)');
legend({'35 Hz', '40 Hz'}, 'Location', 'northwest'); grid on;
title('ASSR magnitude averaged over channels @ 35 Hz and then 40 Hz');

```



From the data, it seems that the 40 Hz ASSR response was higher than the 35 Hz response across every single condition. When the subject was attending to the left ear, the 35 Hz response was **0.0808**, which was slightly smaller than what we saw with the 40 Hz response (at **0.0863**). When the subject was attending to the right ear, we see that the 35 Hz response dropped to **0.0500** while the 40 Hz reached its maximum (at **0.0940**). What is more, during the “attend both” condition, both of the responses weakened (**0.0398** for 35 Hz, and **0.0549** for 40 Hz), and then in the unattended condition we saw that the 35 Hz actually reached its lowest value (**0.0276**) while the 40 Hz remained moderate (at **0.0668**).

As the trials went on, we could see that the 35 Hz magnitude decreases progressively from **attend left --> attend right --> attend both --> unattended**, which could be thought of as showing that the response weakens as the attention shifts away from the left-ear stimulus. However, the 40 Hz response is dominant in every condition, suggesting that there might be an asymmetry in how the auditory system or setup responded to each modulation condition.

Ideally, (and we saw these oddities across all of our homeworks) one would expect the 35 Hz ASSR to be the strongest when attending to the left ear (with the 35 Hz-modulated tone) and the 40 Hz ASSR to be strongest when the subject was attending to the right ear (the 40 Hz-modulated tone). The expected result

would therefore show some sort of frequency-specific enhancement that would be tied to the attended ear, with the unattended frequency decreasing in power.

Instead, the fact that the 40 Hz component manages to win in all of the conditions could be indicative that there is something wrong in the setup or playback. From our brainstorming, Several possible explanations include:

- Unequal stimulus amplitudes between ears (the 40 Hz tone could have been louder, although upon re-listening to our .wav file, the 35Hz tone appears adequately loud).
- Hardware or wiring imbalance, where one channel carried more of both signals (although we have used these earbuds in the past).
- Crosstalk or channel mixing, causing both ears to receive the 40 Hz component.
- Or simply that, to our subject, the 40 Hz is intrinsically more robust and easier for their brain to phase-lock than the 35 Hz.
- Crossing the earbuds and attending to the opposite stimulus for the right and left ears (although through extensive testing we determined this was not the case).

In short, while the data seems to partially reflect attention effects (since the 35 Hz response does weaken as attention moves away), the overall dominance of the 40 Hz component suggests that the results were influenced more by stimulus or recording imbalance than by any sort of true attentional modulation.